

Topic 1: Applied anatomy and physiology

Overview

Students will understand the anatomical/structural and physiological/functional roles performed in the identified systems of the body. They will understand how the controlled stress of exercise will affect the systems and the way that the effect is measured. The topic will cover how different stresses/types of exercise will bring about both acute responses and chronic adaptations.

The principles of Newton's Three Laws of Motion – force, centre of mass and stability – will also be covered and are essential to a student's understanding of how sporting technique and performance can be improved. Students will understand concepts of energy and how they relate to physical activity and sport.

Content

Subject content	What students need to learn																																
1.1 Muscular skeletal system	Knowledge and understanding of the anatomy and physiology of the muscular and skeletal systems. Application of this understanding to physical activity and sport through the content listed below.																																
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	<table><tr><th>Region/ joint</th><th>Muscle(s)</th><th>Bone(s)</th><th>Movement</th></tr><tr><td>Shoulder</td><td>Trapezius Posterior-deltoids Anterior-deltoids Pectoralis Latissimus dorsi</td><td>Humerus Clavicle Scapula</td><td>Horizontal-flexion Horizontal-extension Abduction Adduction Rotation Circumduction</td></tr><tr><td>Hip</td><td>Gluteus Hamstring group Psoas major</td><td>Pelvis Femur</td><td>Flexion Extension Abduction Adduction Rotation Circumduction</td></tr><tr><td>Elbow</td><td>Bicep brachii Tricep brachii</td><td>Radius Ulna Humerus</td><td>Flexion Extension</td></tr><tr><td>Leg and knee</td><td>Quadricep group Hamstring group Gastrocnemius Soleus</td><td>Femur Patella Tibia Fibula</td><td>Flexion Extension</td></tr><tr><td>Ankle and foot</td><td>Gastrocnemius Soleus Tibialis anterior</td><td>Tibia Fibula tarsals Metatarsals Phalanges</td><td>Plantar flexion Dorsi flexion Eversion Inversion</td></tr><tr><td>Wrist and hand</td><td></td><td>Radius Ulna Carpals Metacarpals Phalanges</td><td>Supination Pronation</td></tr><tr><td>Core/trunk</td><td>Rectus Abdominus Latissimus dorsi</td><td>Regions of the vertebral column (cervical, thoracic, lumbar, sacral, coccyx)</td><td>Flexion Extension Rotation</td></tr></table>	Region/ joint	Muscle(s)	Bone(s)	Movement	Shoulder	Trapezius Posterior-deltoids Anterior-deltoids Pectoralis Latissimus dorsi	Humerus Clavicle Scapula	Horizontal-flexion Horizontal-extension Abduction Adduction Rotation Circumduction	Hip	Gluteus Hamstring group Psoas major	Pelvis Femur	Flexion Extension Abduction Adduction Rotation Circumduction	Elbow	Bicep brachii Tricep brachii	Radius Ulna Humerus	Flexion Extension	Leg and knee	Quadricep group Hamstring group Gastrocnemius Soleus	Femur Patella Tibia Fibula	Flexion Extension	Ankle and foot	Gastrocnemius Soleus Tibialis anterior	Tibia Fibula tarsals Metatarsals Phalanges	Plantar flexion Dorsi flexion Eversion Inversion	Wrist and hand		Radius Ulna Carpals Metacarpals Phalanges	Supination Pronation	Core/trunk	Rectus Abdominus Latissimus dorsi	Regions of the vertebral column (cervical, thoracic, lumbar, sacral, coccyx)	Flexion Extension Rotation
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	1.1.2 The stretch-shortening cycle, including the different types of contraction/muscular action: isotonic/eccentric, isotonic/concentric and isometric. Application of how movement or stability is produced as a result of these different contractions/muscular actions during physical activity and sporting movements.
	1.1.3 The concept of agonist, prime mover, antagonist, fixator, synergist and how a muscle can take on these different roles when providing stability or movement in a variety of physical or sporting situations.
	1.1.4 The components of an anatomical lever and how the body uses the lever systems (1st, 2nd and 3rd class) in physical activity and sport. This should include the mechanical advantages and disadvantages of each lever.
	1.1.5 Newton's Three Laws of Motion and how they apply to sporting contexts: Law of Inertia, Law of Acceleration and Law of Action and Reaction.
	1.1.6 The principles related to the stability of the body in relation to the centre of mass and its implication in physical activities.
	1.1.7 The calculation of force and resultant force: a mass of 1 kg exerts a force of 9.81 N (down).
	1.1.8 How the muscular and skeletal systems respond, acutely, both structurally and functionally to the stress of warming up and immediate physical or sporting activity.